

FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Main)-2023

TEST DATE: 12-11-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

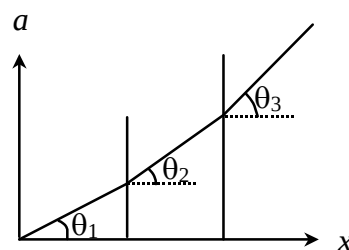
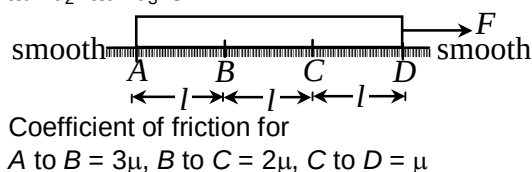
PART – A

SECTION – A

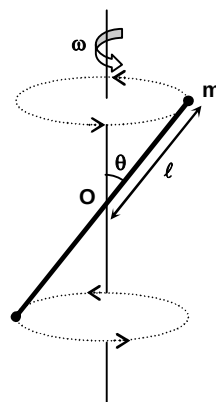
(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In the diagram, the variation of coefficient of friction of the surface is shown. The external force F applied on board is equal to $2\mu Mg$. Now the board is gently pulled so it starts moving. The variation of acceleration of board with respect to displacement has been shown. The ratio $\tan \theta_1 : \tan \theta_2 : \tan \theta_3$ is



- (A) 3 : 2 : 1
 (B) 3 : 4 : 1
 (C) 3 : 4 : 5
 (D) 1 : 2 : 3
2. In terms of the basic units of mass M , length L , time T and charge Q , the dimensions of magnetic permeability of vacuum μ_0 are
 (A) $[MLQ^{-2}]$
 (B) $[ML^2T^{-1}Q^{-2}]$
 (C) $[LTQ^{-1}]$
 (D) $[LT^{-1}Q^{-1}]$
3. A thin massless rod of length 2ℓ has equal point masses m attached at its ends as shown in the figure. The rod is rotating about an axis passing through its centre and making angle θ with it. The magnitude of net torque about O is
 (A) $2m\ell^2\omega^2 \sin\theta \cos\theta$
 (B) $2m\ell^2\omega^2 \sin\theta$
 (C) $2m\ell^2\omega^2 \sin^2\theta$
 (D) $2m\ell^2\omega^2 \cos^2\theta$

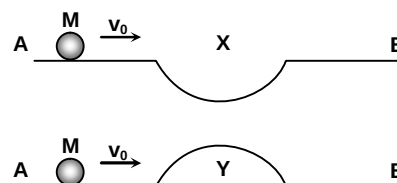


4. A rain drop falling vertically under gravity gathers moisture from the atmosphere at a rate given by $\frac{dm}{dt} = kt^2$, where m is the instantaneous mass, t is time and k is a constant. The equation of motion of the rain drop is $m \frac{dv}{dt} + v \frac{dm}{dt} = mg$
 If the drop starts falling at $t = 0$, with zero initial velocity and initial mass m_0 ($m_0 = 2$ gm, $k = 12$ gm/s³ and $g = 1000$ cm/s²), the velocity v of the drop after one second is
 (A) 250 cm/s
 (B) 500 cm/s
 (C) 750 cm/s
 (D) 1000 cm/s

5. A particle is released at $x = 1$ in a force field $\vec{F}(x) = \left(\frac{2}{x^2} - \frac{x^2}{2} \right) \hat{x}$, $x \geq 0$. Which one of the following statement is FALSE?

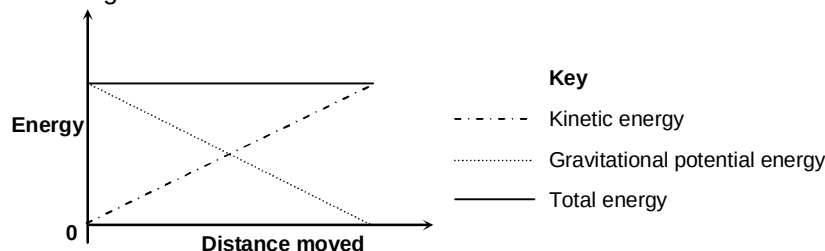
(A) $\vec{F}(x)$ is conservative
 (B) The angular momentum of the particle about the origin is constant
 (C) The particle moves towards $x = \sqrt{2}$
 (D) The particle moves towards the origin

6. A small mass M , is given an initial velocity v_0 , and it slides from A to B via two possible paths; either down the shallow dip X or over the hump Y, both of which are the same shape but inverted. Friction is to be ignored. Along which path does the mass take the shortest time to slide from A to B?



(A) Via X
 (B) Via Y
 (C) You cannot say
 (D) Same time taken

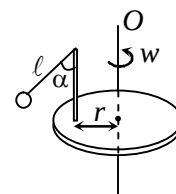
7. The graph below represents the kinetic energy, gravitational potential energy, and total energy of a moving block. Which best describes the motion of the block?



(A) Sliding down an incline with constant friction
 (B) Falling at a constant velocity
 (C) Acceleration on a flat horizontal surface
 (D) Falling freely under gravity

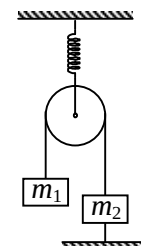
8. A ball of mass m is attached to the end of a thread fastened to the top of a vertical rod which is fitted to a horizontally revolving round table as shown. If the thread forms an angle α with the vertical, the angular velocity ω of table is

(A) $\sqrt{\frac{g}{\ell \cos \alpha}}$
 (B) $\sqrt{\frac{r}{g \tan \alpha}}$
 (C) $\sqrt{\frac{g \tan \alpha}{r + \ell \sin \alpha}}$
 (D) $\sqrt{\frac{g \tan \alpha}{r}}$

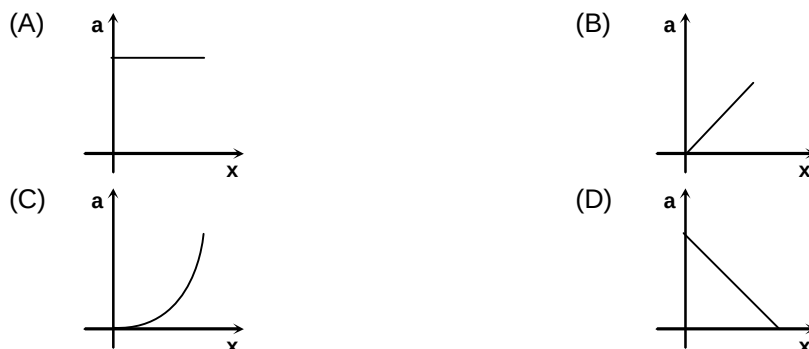
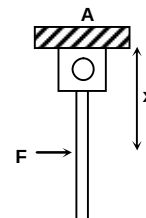


9. In the figure shown, pulley and spring are ideal. If k is spring constant of spring, the potential energy stored in it at equilibrium is ($m_1 > m_2$)

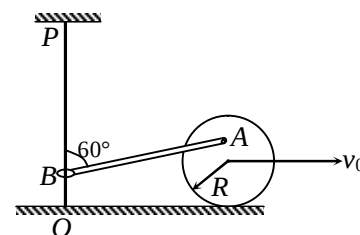
(A) $\frac{2m_1^2 g^2}{k}$
 (B) $\frac{2m_2^2 g^2}{k}$
 (C) $\frac{(m_1 + m_2)^2 g^2}{k}$
 (D) $\frac{1}{2} \frac{(m_1 - m_2)^2 g^2}{k}$



10. A rod of mass m and length ℓ is hinged at one of its end A as shown in the figure. A force F is applied at a distance ' x ' from A. The acceleration of centre of mass ' a ' varies with ' x ' as

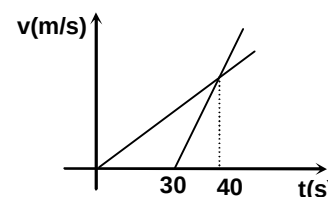


11. A smooth rod OP is fixed vertically. A disc of mass m and radius R is rolling without slipping on a rough horizontal surface as shown. Velocity of centre of mass of disc is v_0 . A rod AB of length L is connected with disc at A, $R/2$ distance vertically up from centre of disc with pin joint (about which it can rotate freely), other end of the rod is attached to a small smooth ring B, which can move freely over rod OP . At an instant rod AB makes an angle 60° with the vertical. The velocity of ring on the vertical rod at this instant is



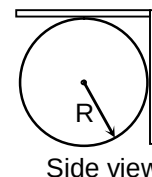
- (A) $\frac{3\sqrt{3}}{2}v_0$ (B) $\frac{3}{2}v_0$
(C) v_0 (D) $\frac{v_0}{2}$

12. The police is chasing a thief and both move on same straight road. After passing a particular point, their velocity v vary with time t as shown in the figure. When will the police catch the thief?
- (A) 90 sec (B) 80 sec
(C) 60 sec (D) 120 sec



13. Last mango in the mango tree is located at 30 m horizontal and 40 m vertical from a boy. The speed and angle with horizontal with which the boy should throw a stone so as to hit the last mango?
- (A) 50 m/s, $\tan^{-1}(2)$ (B) 30 m/s, $\tan^{-1}(3)$
(C) 30 m/s, $\tan^{-1}(2)$ (D) 20 m/s, $\tan^{-1}(3)$

14. A plate bent at right angles along its centre line, is placed onto a horizontal fixed cylinder of radius R as shown. The minimum co-efficient of friction between the cylinder and the plate so that plate does not slip off the cylinder will be (Assume mass of plate and cylinder to be same)



- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$
(C) $\sqrt{2} - 1$ (D) $1 - \frac{1}{\sqrt{2}}$

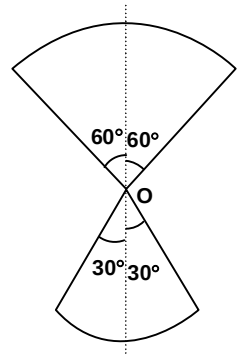
15. Two sectors made of same material and same radius R but angle 120° and 60° are arranged as shown with centre as origin. Find the location of centre of mass?

(A) $\frac{2R(\sqrt{3}-1)}{3}$

(B) $\frac{R(\sqrt{3}-1)}{3}$

(C) $\frac{3R(\sqrt{3}-1)}{3}$

(D) $\frac{4R(\sqrt{3}-1)}{3}$



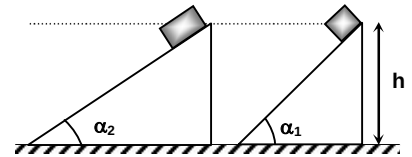
16. Two object of the same mass slide down on two different smooth fixed incline plane of same height but of different angles. When they reach the bottom then I_1 and I_2 are the impulse of the weight of first and second object. Choose the correct option.

(A) $\frac{I_1}{I_2} = \frac{\sin \alpha_1}{\sin \alpha_2}$

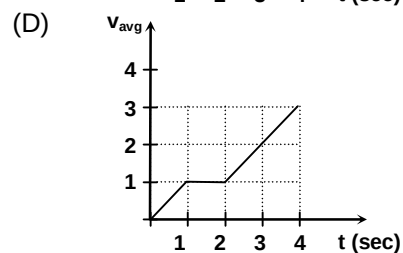
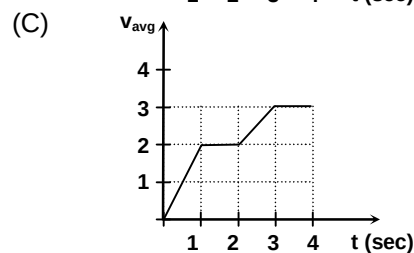
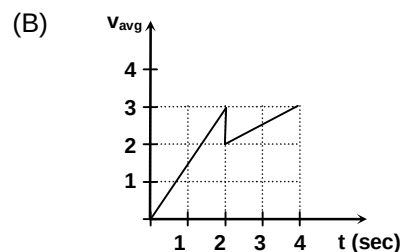
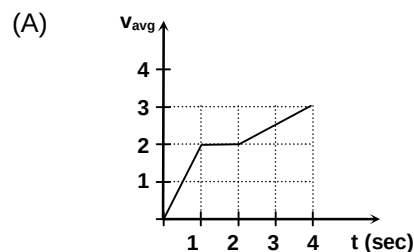
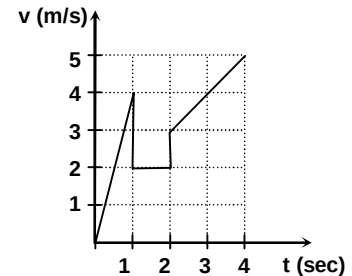
(B) $\frac{I_1}{I_2} = \frac{\sin \alpha_2}{\sin \alpha_1}$

(C) $\frac{I_1}{I_2} = \frac{\tan \alpha_1}{\tan \alpha_2}$

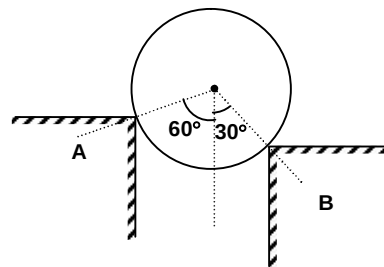
(D) $\frac{I_1}{I_2} = \frac{\tan \alpha_2}{\tan \alpha_1}$



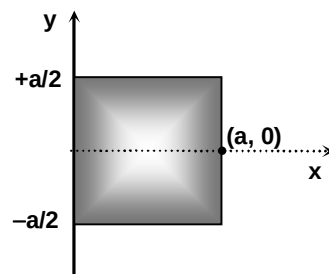
17. The instantaneous velocity of a body moving in a straight line changes with time as shown in the figure. Choose the correct average velocity versus time depicting the same motion?



18. A cylinder of mass M and radius R is resting on two corner edges A and B as shown in the figure. The normal reaction at the edges A and B are: (Neglect friction)



- (A) $N_A = \sqrt{2}N_B$ (B) $N_B = \sqrt{2}N_A$
 (C) $N_A = \frac{Mg}{2}$ (D) $N_B = \frac{2\sqrt{3}Mg}{5}$
19. A square laminar sheet with side a and mass M , has mass per unit area given by $\sigma(x) = \sigma_0 \left[1 - \frac{x}{a} \right]$. Moment of inertia of the sheet about y axis is



- (A) $\frac{Ma^2}{2}$ (B) $\frac{Ma^2}{4}$
 (C) $\frac{Ma^2}{6}$ (D) $\frac{Ma^2}{12}$
20. A spring is compressed between two toy carts of masses m_1 and m_2 . When the toy carts are released, the spring exerts on each toy cart equal and opposite forces for the same time t . If the coefficients of friction μ between the ground and the toy carts are equal, then the displacements of the toy carts are in the ratio
- (A) $\frac{s_1}{s_2} = \frac{m_2}{m_1}$ (B) $\frac{s_1}{s_2} = \frac{m_1}{m_2}$
 (C) $\frac{s_1}{s_2} = \left(\frac{m_2}{m_1} \right)^2$ (D) $\frac{s_1}{s_2} = \left(\frac{m_1}{m_2} \right)^2$

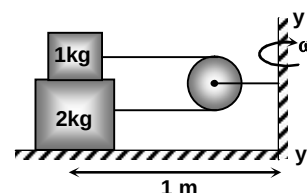
SECTION – B

(Numerical Answer Type)

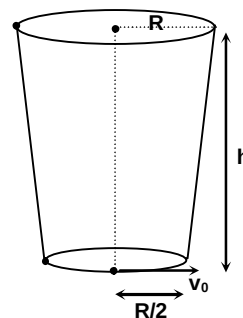
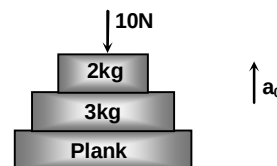
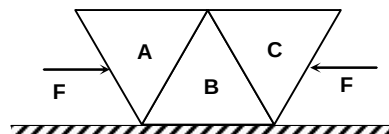
This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A shell is fired at an angle θ with horizontal up a plane inclined at an angle $\frac{\pi}{6}$. The value of θ for maximum range is $\frac{\pi}{k}$. Find the value of k .

22. Two blocks A and B 1 kg and 2 kg respectively are kept on a turn table 1 m away from axis yy' as shown in the figure. Consider $\mu = 0.5$ between all contact surfaces and pulley to be massless and frictionless. The angular speed of the turn table for which the blocks just begin to slide is $\sqrt{5k}$. Find k . (Take $g = 10 \text{ m/s}^2$)



23. Two objects of masses m and $2m$ are moving at speeds of v and $v/2$, respectively. After undergoing a completely inelastic collision, they move together with a speed of $v/3$. The angle between the initial velocity of the two objects is K° . Find the value of k .
24. If there is a positive error of 50% in the measurement of velocity of a body, then the error in the measurement of kinetic energy is $k\%$. Find the value of k .
25. The motion of a point A on the rim of a wheel rolling uniformly along a horizontal path has the form
 $x = 2(5t - \sin 5t)$
 $y = 2(1 - \cos 5t)$
 The distance which it traverses between two successive contacts with the roadbed is k meter. Find the value of k .
26. Three identical equilateral prisms A, B and C are arranged as shown in the figure. Mass of each prism is $M = \sqrt{3}$ kg. Two horizontal forces of equal magnitude (F) are applied on A and C as shown in the figure. Find the minimum value of F in newton so that A and C loose contact with the floor. (Ignore friction and take $g = 10 \text{ m/s}^2$)
27. A modern machine gun fires 3600 bullets per minute with 600 km/hr. If each bullet weights 200 gm then the power of the gun is $\frac{500}{m}$ kilowatt. Find the value of m .
28. A solid cylinder rolls down an incline plane of angle θ . The least value of coefficient of friction so that the cylinder rolls without slipping is $\frac{\tan \theta}{k}$. Find the value of k .
29. Two blocks of masses 2 kg and 3 kg are placed on a plank. An external force of 10 N is acting on 2 kg block and the whole system including the plank is moving vertically up with a constant acceleration $a_0 = 2 \text{ m/s}^2$. The normal reaction force between the 3 kg block and Plank is α N. Find the value of $\frac{\alpha}{10}$. ($g = 10 \text{ m/s}^2$)
30. A smooth cylindrical glass lies on the ground. A small particle of mass m lies on the bottom surface touching circumference as shown in the figure. The minimum value of v_0 (along the circumference) such that particle just reaches at the top of glass is $\sqrt{\frac{Kgh}{3}}$. Find the value of K .



Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

31. The CORRECT order of electron gain enthalpy of B, Be, N and C is (consider magnitude only)
 (A) $\text{Be} < \text{B} < \text{C} < \text{N}$
 (B) $\text{Be} < \text{N} < \text{B} < \text{C}$
 (C) $\text{N} < \text{Be} < \text{C} < \text{B}$
 (D) $\text{N} < \text{C} < \text{B} < \text{Be}$
32. Which of the following properties is INCORRECTLY matched?
 (A) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ (Oxidising character)
 (B) $\text{K} > \text{Mg} > \text{Al} > \text{B}$ (Metallic character)
 (C) $\text{C} > \text{Be} > \text{B} > \text{Li}$ (1^{st} Ionisation enthalpy)
 (D) $\text{Li} > \text{Na} > \text{K} > \text{Rb}$ (Chemical reactivity)
33. The minimum uncertainty in the velocity of a particle of mass $1.1 \times 10^{-28} \text{ kg}$ will be if its uncertainty in position is $3 \times 10^{-10} \text{ cm}$ ($h = 6.62 \times 10^{-34} \text{ Js}$).
 (A) $1.6 \times 10^6 \text{ ms}^{-1}$
 (B) $2.6 \times 10^5 \text{ ms}^{-1}$
 (C) $3.6 \times 10^5 \text{ ms}^{-1}$
 (D) $1.6 \times 10^5 \text{ ms}^{-1}$
34. The coefficients of BaCrO_4 , KI , CrCl_3 and KCl in the balanced equation of the following reaction are respectively
 $\text{BaCrO}_4 + \text{KI} + \text{HCl} \longrightarrow \text{BaCl}_2 + \text{I}_2 + \text{KCl} + \text{CrCl}_3 + \text{H}_2\text{O}$
- | | BaCrO_4 | KI | CrCl_3 | KCl |
|-----|------------------|-------------|-----------------|--------------|
| (A) | 2 | 6 | 2 | 6 |
| (B) | 6 | 2 | 6 | 6 |
| (C) | 2 | 2 | 6 | 6 |
| (D) | 6 | 6 | 2 | 2 |
35. At a given temperature, the rate constant for the decomposition of HI on a metal surface is 0.06 Ms^{-1}
 $2\text{HI}(\text{g}) \longrightarrow \text{H}_2(\text{g}) + \text{I}_2(\text{g})$
 The time taken for the concentration of HI to fall from 1.50 M to 0.30 M is
 (A) 10 s
 (B) 20 s
 (C) 15 s
 (D) 8 s

36. The increasing order of bond angles in OF_2 , OCl_2 , ClO_2 and H_2O is.....
 (A) $\text{OF}_2 < \text{H}_2\text{O} < \text{OCl}_2 < \text{ClO}_2$
 (B) $\text{H}_2\text{O} < \text{F}_2\text{O} < \text{Cl}_2\text{O} < \text{ClO}_2$
 (C) $\text{OF}_2 < \text{Cl}_2\text{O} < \text{H}_2\text{O} < \text{ClO}_2$
 (D) $\text{ClO}_2 < \text{Cl}_2\text{O} < \text{OF}_2 < \text{H}_2\text{O}$
37. Consider the following reactions in equilibrium

$$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3 \quad (K_1)$$

$$\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO} \quad (K_2)$$

$$\text{H}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{H}_2\text{O} \quad (K_3)$$
 Then the equilibrium constant for the reaction

$$2\text{NH}_3 + \frac{5}{2}\text{O}_2 \rightleftharpoons 2\text{NO} + 3\text{H}_2\text{O}$$
 (All the above reactions are in homogeneous equilibrium)
 (A) $\frac{K_2 K_3}{K_1}$
 (B) $\frac{K_2 K_1^3}{K_3}$
 (C) $\frac{K_2 K_3^3}{K_1}$
 (D) $\frac{K_2 K_1}{K_3}$
38. The pH of pure water at 25°C and 35°C are 7 and 6 respectively. The heat of formation of H^+ (aq) and OH^- (aq) ions in Kcal mol^{-1} is
 (A) -84.5
 (B) $+84.5$
 (C) -94.5
 (D) $+94.5$
39. A hypothetical small particle of mass 'm' moves in such a way that its potential energy $= -\frac{1}{2} mkr^4$. Where k is a constant and r is its distance from the nucleus (origin). Assuming Bohr's model r is directly proportional to
 (A) n^2
 (B) n
 (C) $\sqrt{n}$
 (D) $\sqrt[3]{n}$

40. Consider the reaction in equilibrium
 $\text{SnO}_2(\text{s}) + 2\text{H}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\text{g}) + \text{Sn}(\ell)$
 at 300 K, the equilibrium mixture was found to contain 40% by volume of $\text{H}_2(\text{g})$. The K_p for the reaction is
 (A) 1.15
 (B) 10.9
 (C) 7.75
 (D) 2.25
41. The three reactions are given below
 $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons \text{C}(\text{g}) + \text{D}(\text{g}) \quad \dots \text{(I)}$
 $\text{A}(\text{g}) \rightleftharpoons \text{C}(\text{g}) + \text{D}(\text{g}) \quad \dots \text{(II)}$
 $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons \text{D}(\text{g}) \quad \dots \text{(III)}$
 Which of the following statement is INCORRECT?
 (A) Increase in pressure shifts the equilibrium (II)nd to the left
 (B) Decrease in volume shifts the (III)rd equilibrium to right
 (C) Addition of He at constant volume shifts the (II)nd equilibrium to left
 (D) Addition of He at constant pressure shifts the (III)rd equilibrium to left
42. At high temperature and at a pressure of around 2 atm, nitrogen gas is bubbled over calcium carbide, the major product obtained is
 (A) Calcium cyanide
 (B) Calcium cyanamide
 (C) Calcium nitride
 (D) Calcium
43. Which of the following statement is NOT correct?
 (A) Stability of peroxides and superoxides of alkali metals increases with increase in the size of metal ion.
 (B) Sodium hydroxide does not absorb water from atmosphere.
 (C) Lithium fluoride is less soluble in water due to its high lattice enthalpy.
 (D) Caesium iodide is less soluble in water due to low hydration enthalpy.
44. In solid state, beryllium chloride has
 (A) Three centre-four electron bonds.
 (B) Three centre-three electron bonds.
 (C) Two centre-three electron bonds.
 (D) Two centre-two electron bonds.
45. Consider the following silicates
 I. Pyro silicates II. Cyclic silicates
 III. Double chain silicates IV. Single chain silicates
 V. 3D-silicates VI. Sheet silicates
 The silicates in which **only two** corners **per tetrahedron** are shared
 (A) I, II, IV
 (B) IV, VI
 (C) I, VI
 (D) II, IV

46. Select the CORRECT sequence of geometry of the given molecules/ions
 Borazon, Borazole, $B_3O_6^{3-}$, Al_2Cl_6
 (A) Non planar, non planar, non planar, planar
 (B) Planar, planar, non planar, non planar
 (C) Non planar, non planar, non planar, planar
 (D) Non planar, planar, planar, non planar
47. Consider the following species
 NO_2^+ , NO_2^- , NO_2 , NO_3^-
 Select the INCORRECT statement for the above molecule/ions?
 (A) NO_2 is paramagnetic
 (B) NO_2^+ is linear, NO_2^- is bent having bond angle more than 120°
 (C) NO_3^- has bond order 1.33
 (D) NO_2 has bond angle more than 120°
48. Consider the compounds LiH, NaH, Li_2CO_3 and Na_2CO_3 . Select the compound (s) which decomposes on heating around $400 - 500^\circ C$.
 (A) LiH, Li_2CO_3
 (B) NaH, Na_2CO_3
 (C) LiH, NaH
 (D) NaH, Li_2CO_3
49. Which of the following statement(s) is(are) NOT correct?
 I. Bond length order is $H_2^- = H_2^+ > H_2$.
 II. O_2^+ , NO , N_2^- have bond order 2.5.
 III. Bond order varies from 0 to 4.
 IV. In NO_3^- and BO_3^- the bond order is same between central atom and side atom.
 (A) I, II
 (B) II, III
 (C) I, IV
 (D) I, III
50. Select the INCORRECT statement regarding Diborane (B_2H_6)
 (A) Maximum 6 H-atoms lie in a plane
 (B) Maximum 6 atoms lie in a plane
 (C) There is hindered rotation between two boron atoms
 (D) Terminal H – B – H bond angle is greater than bridge H – B – H bond angle

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. A sample of solid AgCl was treated with 5.00 mL of 1.5 M Na_2CO_3 solution to give Ag_2CO_3 . The remaining Cl^- ion concentration was found to be 0.0026 g L^{-1} . The solubility product of AgCl is given by $x \times 10^{-y}$. Then the value of (x + y) is.....(Rounded off to nearest integer).
 (K_{SP} of $Ag_2CO_3 = 8.2 \times 10^{-12}$)

52. A tiny particle moving with a velocity $\frac{1}{60}$ th times the velocity of light. If uncertainty in the position is $\frac{6.62}{3.14} \times 10^{-12}$ m, then the minimum uncertainty in kinetic energy is $x \times 10^{-y}$ J, then the value of xy is.....
53. Potassium permanganate reacts with sulfur dioxide in acidic medium to produce a product containing bisulphate ion(s) and same other product(s). The sum total of coefficients of potassium permanganate and the product containing bisulphate ions in the balanced equation is.....
54. For a reaction which following first order kinetics the half life is 693 s and the values of pre exponential factor and activation energy are 4×10^{13} Hz and 98.6 kJ mol^{-1} respectively, then the temperature in Kelvin when half life is 693 s, is..... ($\log 2 = 0.30$)
55. An equilibrium mixture at 300 K contains N_2O_4 and NO_2 at 0.28 and 1.1 atm respectively. If the volume of container get doubled, the equilibrium pressure of NO_2 (g) is $x \times 10^{-2}$ atm, then the value of x is.....
56. Total number of species having used all three P-orbitals for hybridisation of central atom and are non-polar also is.....
 XeO_2F_2 , SnCl_2 , IF_5 , I_3^+ , XeO_4 , $[\text{XeF}_7]^+$, SO_2
57. The number of correct order(s) in the following is.....
 1. $\text{CH}_4 < \text{CCl}_4 < \text{CF}_4$ (Electronegativity of carbon)
 2. $\text{Mg}^{++} < \text{K}^+ < \text{S}^{--} < \text{Se}^{--}$ (Ionic radius order)
 3. $\text{LiF} > \text{NaF} > \text{KF} > \text{RbF}$ (Lattice energy order)
 4. $\text{Li}^+ < \text{Mg}^{++} < \text{Al}^{3+}$ (Hydration energy order)
 5. $\text{Cl}_2 > \text{F}_2 > \text{Br}_2 > \text{I}_2$ (Bond energy order)
58.
$$\begin{array}{c} \text{Na}_2\text{O}_2 \xrightarrow[\text{(ice cold)}]{\text{H}_2\text{O}} \text{P} + \text{Q} \\ \downarrow \text{H}_2\text{O (Room temperature)} \\ \text{P} + \text{R} \uparrow \end{array}$$

 If molar mass of (Q) and (R) are M_1 and M_2 respectively then the value of $\left(\frac{M_1 + M_2}{2}\right)$ is.....
59. A 1.025 g sample containing a weak acid HA (MW = 82) is dissolved in 60 mL of water and titrated with 0.25 M NaOH. When half of the acid was neutralized, the pH was found to be 5.0 and at the equivalence point the pH is 9.0. The mass percent of weak acid HX in the sample is.....%.
60. The pH of the resultant solution of 25 mL of 0.1 M HCl, 50 mL of 0.2 M HNO_3 and 25 mL of 0.1 M NaOH is.....

Mathematics**PART – C****SECTION – A****(Single Choice Answer Type)**

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

61. Let f be a differentiable function satisfying $\int_0^x (x-p+1)f(p)dp = x^4 + x^2 \forall x \in [-2, \infty)$, then value of $\int_{-2}^2 (f'(x) + f(x))dx$ is
 (A) 0 (B) 36
 (C) 72 (D) 18
62. Let $P = \lim_{x \rightarrow 0} \frac{1 - (\cos 2x \cos 4x \cos 6x \cos 8x \cos 10x)^3}{(\tan^2 x)}$, then the value of $\left(\frac{P}{10}\right)$ is
 (A) 66 (B) 33
 (C) 22 (D) 11
63. If $A = \lim_{n \rightarrow \infty} \sum_{r=1}^n \sin\left(\frac{r}{n^2}\right)$ and $B = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sin\left(\frac{\pi r}{n}\right)$, then value of $2A + B$ is
 (A) $1 + \frac{2}{\pi}$ (B) $\frac{2}{\pi}$
 (C) $\frac{\pi-1}{2}$ (D) $\frac{\pi}{2} - 1$
64. Let $f(x) = \frac{1}{\sin^2 x - (x-a)^2}$, $x \in (0, 1)$, then which of the following is **INCORRECT** ?
 (A) $f(x)$ is always increasing in $(0, 1)$ (B) $f(x)$ is always decreasing in $(0, 1)$
 (C) $f(x)$ is increasing if $a \in (2, 3)$ (D) $f(x)$ is decreasing if $a \in (2, 3)$
65. If $A = \int_0^1 (1-x^{2023})^{\frac{1}{2022}} dx$ and $B = \int_0^1 (1-x^{2022})^{\frac{1}{2023}} dx$, then find the value of $\left(\frac{A^3 - B^3}{B^3}\right) + \left(\frac{A^4 + B^4}{B^4}\right)$ is
 (A) 0 (B) 1
 (C) 2 (D) none of these
66. The exhaustive set of domain of the function $f(x) = \operatorname{sgn}\left(\frac{x^2 + x + 4}{x^2 + x + 3}\right) + \tan\left(\left[\frac{x^2 + x + 3}{x^2 + x + 4}\right]\right)$, (where $[.]$ is G.I.F.) is
 (A) $(-\infty, 0)$ (B) $\left(\frac{1}{3}, 3\right)$
 (C) $\left[\frac{1}{3}, 3\right]$ (D) $(-\infty, \infty)$

67. If the area bounded between the curves $y = \tan x$, $y = \sin x$, $x = 0$ and $x = \frac{\pi}{4}$ is A, the A is equal to
- (A) $\frac{1}{2} \ln 2 + \frac{1}{\sqrt{2}}$ (B) $\frac{1}{2} \ln 2$
 (C) $\frac{1}{\sqrt{2}}$ (D) none of these
68. Let $f(x)$ be a function, such that $f(x) = x + \sin x$ and $I_1 = \int_0^{\pi} f(x) dx$, $I_2 = \int_0^{\pi} f^{-1}(x) dx$ and $I_3 = \int_0^1 e^{-x^2} dx$, then which of the following is true ?
- (A) $I_1 > I_2 > I_3$ (B) $I_3 > I_2 > I_1$
 (C) $I_1 > I_3 > I_2$ (D) $I_3 > I_1 > I_2$
69. If $f(x) = \int_0^x (\ln t)^2 dt$ and $g(x) = \int_0^x (\ln t) dt$, then area bounded between the curve $y = f'(x)$ and $y = g'(x)$ between $x = 1$ to $x = e$ is
- (A) $e - 3$ (B) 3
 (C) e (D) $3 - e$
70. Let $L_1 = \lim_{x \rightarrow \infty} \sec^{-1} x$, $L_2 = \lim_{x \rightarrow -\infty} \sec^{-1} x$, then the value of $\int_{L_1}^{L_2} \frac{x^3 + 1}{x^2 + 1} dx$
- (A) π (B) $\frac{\pi}{2}$
 (C) $\frac{\pi}{4}$ (D) 0
71. $\lim_{x \rightarrow 1} \left(\sum_{n=1}^{12} \frac{x}{n(n+1)x^2 + 3(2n+1)x + 9} \right)$ is equal to
- (A) $\frac{11}{60}$ (B) $\frac{3}{16}$
 (C) $\frac{13}{64}$ (D) $\frac{9}{44}$
72. Water is being filled at the rate of 100 litre/min in a conical vessel (vertex downwards) with axis vertical. Its semi vertical angle is $\tan^{-1}\left(\frac{5}{12}\right)$. The rate (in square meter per minute) at which the wet curved surface area of the tank is increasing, when the depth of water in tank is 3 meters is
- (A) $\frac{13}{75}$ (B) $\frac{13}{25}$
 (C) $\frac{1}{5}$ (D) $\frac{3}{11}$
73. The area of the region bounded by $A = \{(x, y) : x^2 - 4 \leq y \leq \min\{2x + 11, 9, 12 - 6x\}\}$ is
- (A) $\frac{511}{12}$ (B) $\frac{947}{12}$
 (C) $\frac{437}{8}$ (D) $\frac{240}{7}$

74. Let $f(t) = \int_0^t e^{x^2} \left(\frac{x^5}{(x^4 + 2x^2 + 2)^2} \right) dx$, then $f(1) + f'(1)$ is equal to
- (A) $\frac{3e}{10} - \frac{1}{4}$ (B) $\frac{7e}{50} - \frac{1}{4}$
 (C) $\frac{7e}{50} - \frac{1}{2}$ (D) $\frac{2e}{5} - \frac{1}{2}$
75. $\lim_{x \rightarrow \infty} \left(\sqrt{x^4 + kx^3 + 3x^2 + mx + 2} - \sqrt{x^4 + 2x^3 - nx^2 + 3x - p} \right) = 4$ then $k + n$ is equal to
- (A) 3 (B) 7
 (C) 5 (D) 0
76. If $I = \int_0^{\sqrt{2\pi}} \frac{x^3 \sin^{2022} x^2}{\cos^{2022} x^2 + \sin^{2022} x^2} dx$ is equal to
- (A) $\frac{\pi^2}{2}$ (B) $2\pi^2$
 (C) π^2 (D) 0
77. If the solution curve of the differential equation $\frac{dy}{dx} = \frac{x+y-4}{x-y}$ passes through the point (3, 2) and $(p+2, 3)$, $p > 0$, then
- (A) $2 \tan^{-1} \left(\frac{1}{p} \right) = \log_e (p^2 + 1)$ (B) $\tan^{-1} \left(\frac{1}{p} \right) = \log_e (p^2 + 1)$
 (C) $2 \tan^{-1} \left(\frac{1}{p+1} \right) = \log_e (p^2 + 2p + 2)$ (D) $2 \tan^{-1} \left(\frac{1}{p} \right) = \log_e \left(\frac{p^2 + 1}{p} \right)$
78. Let $f : [-4, 2] \rightarrow \mathbb{R}$ be given as $f(x) = \begin{cases} \min\{(x+6), x^2\} & -4 \leq x \leq 0 \\ \max\{\sqrt{x}, x^2\} & 0 \leq x \leq 2 \end{cases}$
- If the area bounded by $y = f(x)$ and x -axis in A then the value of $3A$ is equal to
- (A) 19 (B) 35
 (C) 37 (D) 32
79. If $f(t) = \int_1^t \frac{\log_{10} x}{1+x} dx$, $t > 0$, then $f(e^4) + f(e^{-4})$ is equal to
- (A) 8 (B) $\frac{16}{\log_e 10}$
 (C) $\frac{8}{\log_e 10}$ (D) $\frac{e^4 - 1}{2 \log_e 10}$
80. Let $f(x) = \begin{cases} x^3 - 2x^2 + 3x - 9 & x \leq 2 \\ -3x + \log_3(k^2 - 5) & x > 2 \end{cases}$, then the set of all values of k for which $f(x)$ has maximum value at $x = 2$ is
- (A) $[-4\sqrt{2}, 4\sqrt{2}]$ (B) $(-\sqrt{5}, 4\sqrt{2})$
 (C) $[-4\sqrt{2}, -\sqrt{5}) \cup (\sqrt{5}, 4\sqrt{2}]$ (D) $(\sqrt{5}, 2\sqrt{2}]$

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. The number of solution of the equation $\sin^{-1}(x) = \left[\frac{x^2 + x + 2}{x^2 + x + 1} \right] = 0$, [.] is G.I.F.
82. The area bounded between the curves $y = \sin^{-1}(\sin x)$ and $y = \sqrt{\frac{\pi^2}{4} - \left(\pi - \frac{\pi}{2}\right)^2}$ is 'A', then the value of $\left[\frac{8A}{\pi^2} \right]$ (where [.] is G.I.F) is
83. Let $f(x)$ be a function defined as $f(x) = \begin{vmatrix} 0 & -x & -x^2 \\ x & 0 & x+1 \\ x^2 & -x-1 & 0 \end{vmatrix} + \left[\frac{x}{2} \right] \sin x + \sqrt{3(4-x^2)}$, then the number of integers in the range of $f(x)$ is (where [.] is G.I.F)
84. The number of points where the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x - 2| \cos|x^2 - 3x| \sin|x - 2| + (x - 1)|x^3 - 8x^2 + 7x|$ is non differentiable is
85. Let $m = \begin{bmatrix} a & 0 & 1 \\ a & b & -b \\ 0 & 1 & c \end{bmatrix}$ and $\det(m) = 37$, if a, b, c are positive integers, then least value of $\text{trace}(m)$ is
86. If $y = y(x)$ is an implicit function of x such that $\log_e(x + y) = 6xy$, then $\frac{d^2y}{dx^2}$ at $x = 0$ is equal to _____
87. If the value of $\lim_{x \rightarrow 0} \left(3 - 2\cos x \sqrt{\cos 2x} \right)^{\left(\frac{x+3}{x^2} \right)}$ is equal to e^a , then a is equal to
88. If the tangent to the curve $y = x^3 - 2x^2 + 4x + 4$, at the point (a, b) (where $a > 0$) is also tangent to the curve $y = 3x^2 + 2x - 1$ at $(1, 4)$, then $a + b$ is equal to
89. If $f(x) = \int \frac{12x^{15} + 5x^{18}}{(2x^3 + 3 - 7x^8)^3} dx$ and $16f(1) - 1 = 0$ and $f(-1) = \frac{1}{k}$, then $|k| =$
90. Let $I_n = \int_1^e x^{23} (\log|x|)^n dx$ where $n \in \mathbb{N}$, if $24I_{15} = \alpha I_{14} + \beta I_{13}$, then $\alpha + \beta$ is equal to